Jan 15, 2022

AGGIE INVITATIONAL

Dynamic Planet C

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- You will be given 50 minutes to complete this test
- One 2" or smaller three-ring binder and two calculators are permitted per team
- Point values are labeled either at section headers or in parentheses after question.
- **DO NOT** write on the test
- Be clear, concise, and specific with your answers
- Make sure your team number is on every page

Good Luck!

Section I: Multiple Choice

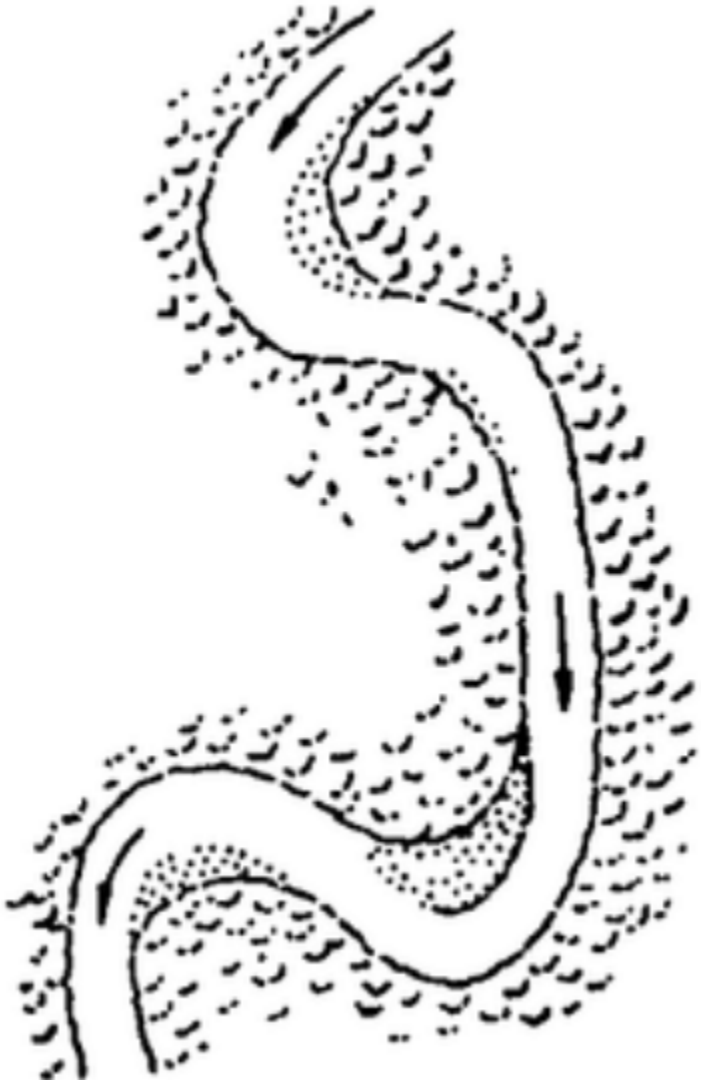
1. Where is a majority of Earth's freshwater found?
   A) Polar Ice Caps/Glacial Ice
   B) Aquifers
   C) Oceans
   D) Rivers
   E) Lakes

2. Approximately what percent of the Earth's water supply is found in freshwater resources that are groundwater deposits (i.e. rivers, lakes, and streams)?
   A) 20%
   B) 2%
   C) 0.2%
   D) 0.02%
   E) 0.002%

3. A rainstorm causes a recharge of 1 inch in an aquifer of clean, well-sorted sand. Estimate the change in water table elevation.
   A) 2.86 - 5 feet
   B) 0.28-0.5 inches
   C) 2.86 - 5 inches
   D) 3.33 - 6.67 inches
   E) 3.33 - 6.67 feet

4. The volumetric flux of a poorly sorted sand aquifer is 0.25m/day. The effective porosity is 0.5. How long will it take for the water to travel 100m?
   A) 20 days
   B) 200 days
   C) 2000 days
   D) 2.0 days
   E) 0.2 days

5. Describe the type of river channel present in the following image:



A) Sinuous uniform channel
B) Sinuous point-bar channel
C) Braided point-bar channel
D) Bar-braided channel
E) Straight Channel

6. If the distance between two wells is 100 meters, and the piezometric surface drop between two wells is 4 meters, find the time it would take for a molecule of water to move from one well to the other. Assume steady unidirectional flow in a homogenous sand confined aquifer with hydraulic conductivity of K = 2 m/day and an effective porosity of 0.40.

A) 1.037 years
B) 0.91 years

C) 1.37 years
D) 70.4 years
E) 7.04 years

7. Which of the following would be an indication that a body of water was contaminated by human waste water?

A) low BOD and a fecal coliform bacteria count of zero
B) high levels of nutrients, such as nitrogen and phosphorous, and high BOD
C) low BOD and low levels of nutrients, such as nitrogen and phosphorus
D) low levels of nutrients, such as nitrogen and phosphorous, and a fecal coliform bacteria
E) High BOD, and low levels of nutrients, such as nitrogen and phosphorus

8. Acid mine drainage results from acidic water formed belowground that makes its way to the surface; the acidic water is formed as a result of the flooding of abandoned mines where the underground water

A) reacts with a type of rock, pyrite, which releases iron and hydrogen ions
B) reacts with sulfur dioxide and nitrogen dioxide to form sulfuric and nitric acids
C) flushes out the chemicals used in the mining process
D) permeates a limestone layer that lowers the pH
E) reacts with copper and aluminum to form pyrite rock and hydrogen ions

9. Acidification of lake and reservoir surface waters is caused by what kind of pollution?

A) Air and water pollutants
B) Water pollutants only
C) Soil and Light pollutants only
D) Soil, Water, or Air pollutants.
E) Air pollutants only

10. What is Kegelkarst?
    A) type of tropical karst terrain with numerous cone-like hills
    B) karstic landscape which is developed beneath a cover of insoluble rocks
    C) a deep sinkhole resulting from collapse of limestone bedrock that exposes groundwater underneath
    D) an inverted funnel-shaped sinkhole
    E) a spring emerging from karst, originating a flow of water on the surface

11. Given that the radius of a solid riser pipe is 0.413 meters, and the screen length is 4.413 meters, find the hydraulic conductivity. Assume t_o is 1.

    A) 0.458 m/s
    B) 4.58 m/s
    C) 0.0458 m/s
    D) 0.386 m/s
    E) 3.86 m/s

12. What percentage of erosion is human accelerated?
    A) 40%
    B) 50%
    C) 60%
    D) 70%
    E) 80%
13. Which are common pollutants in groundwater aquifers?
    A) Arsenic, hydrogen sulfide, Radon
    B) Radon, Oxygen, Nitrogen
    C) Nitrogen, Oxygen, Helium
    D) Hydrogen sulfide, Helium, Nitrogen
    E) Iron Oxide, Arsenic, Elemental Hydrogen

14. What is the most widespread type of karst in the world?
    A) Karren
    B) Sinkhole
    C) Poljes
    D) Glaciers
    E) Caves

15. If the distance between two wells is 25 meters, and the piezometric surface drop between two wells is 7.2 meters, find the time it would take for a molecule of water to move from one well to the other. Assume steady unidirectional flow in a homogenous sand confined aquifer with

hydraulic conductivity of K = 0.25 m/day and an effective porosity of 0.61. Round to the tenth place for your final answer.

A) 212.1 days
B) 212.2 days
C) 211.8 days
D) 211.1 days
E) 210.1 days

16. What is the SI unit for flow rate?

A) m^2/s
B) m^3/s
C) m/s
D) m/min
E) m^3/min

17. Which of the following is a term used to describe the open surface waters in a lake, away from the shore?

A) Benthic Zone
B) Limnetic Zone
C) Eutrophication
D) Littoral Zone
E) Nekton

18. Precipitation that carries pollutants from land into bodies of water is:

A) Runoff
B) Eutrophication
C) Rain
D) Condensation
E) Sleet

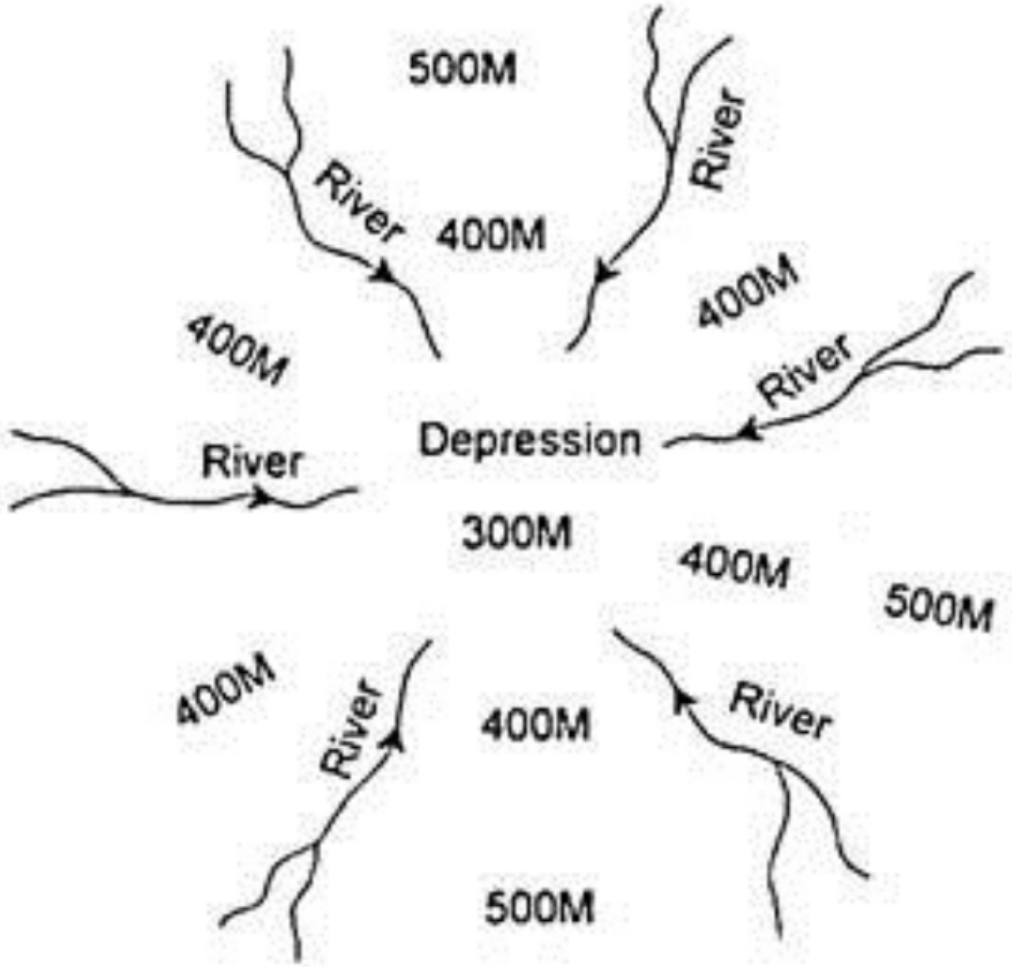
19. Which of the following is a shallow zone in a freshwater habitat where light reaches the bottom and nurtures plants?

A) Benthic Zone
B) Limnetic Zone
C) Littoral Zone
D) Benthos
E) Profundal Zone

20. A drainage pattern in a volcanic environment is most likely to be ____ drainage.

A) Trellised
B) Rectangular
C) Radial
D) Barbed
E) Pinnate

21. True or False: The world's largest limestone karst is in Mexico's Chiapas. \
    A) True
    B) False
22. The following image is an example of what kind of drainage pattern?



    A) Radial
    B) Barbed
    C) Annular
    D) Trellised
    E) Centripetal
23. What are the types of water flow in soil?
    A) Turbulent Flow and Laminar Flow
    B) Linear Flow only
    C) Laminar Flow Only
    D) All of the above
    E) None of the above

24. In what way do fluid particles flow in a turbulent flow?
    A) Twisting
    B) Crossing
    C) Re-Crossing
    D) Definite Path
    E) None of the Above
25. Which of the following equations defines Darcy's law?
    A) $q = KA$
    B) $q= K/A$
    C) $q = k+a$
    D) $q= KiA$
    E) $q = Ki/A$
26. The coefficient of permeability in fine sand is which of the following?
    A) 1.0 and greater
    B) $1*10^{-6}$ and smaller
    C) $1*10^{-3}$ to $5*10^{-2}$
    D) $5*10^{-3}$ to $1*10^{-4}$
    E) $5*10^{-4}$ to $1*10^{-3}$
27. Darcy's Law is only valid if it is equal to or less than:
    A) 0.5
    B) 0.25
    C) 1
    D) 0.1
    E) 100
28. Darcy's Law is only valid for:
    A) Turbulent Flow
    B) Turbulent Flow and Hydraulic Flow
    C) Laminar Flow
    D) Laminar FLow and Hydraulic Flow
    E) Laminar Flow and Turbulent Flow
29. The value of the specific yield (SY) of an aquifer depends on:
    A) Grain Size
    B) Compaction of Stratum
    C) Grain Shape
    D) All of the Above
    E) None of the Above
30. In most confined aquifers, the value of the storage coefficient ranges from:
    A) 0.05 to 0.0005
    B) 0.5 to 0.005
    C) 0.005 to 0.0005

D) 0 to 5
E) 0.0000005 to .0000005

31. Darcy's Law is not valid under which circumstance:
    A) At Well Face
    B) Top Surface of Well
    C) Bottom Surface of Well
    D) Sides of Well
    E) All of the above
32. What lake stratum is commonly known as the thermocline?
    A) Metalimnion
    B) Epilimnion
    C) Hypolimnion
33. Which area of a lake comprises the pelagic zone?
    A) Open water of lake
    B) Shallow perimeter of lake
    C) Shoreline of lake
    D) Deep middle of lake
    E) None of the above
34. What is a seiche?
    A) Another term for storm surge
    B) A standing wave oscillating in an enclosed body of water
    C) Disk used to measure water quality
    D) Another term for tsunami
    E) None of the above
35. When is a lake's turnover most likely to occur?
    A) Spring
    B) Winter
    C) Summer
    D) Fall
36. Where is most methane emitted by lakes produced?
    A) Lake sediments when oxygen is no longer present
    B) Thermocline
    C) Epilimnion
    D) Hypolimnion
    E) Lakes don't emit methane :)
37. As a whole, are lakes net sinks of CO_2 or net sources of CO_2?
    A) Sinks
    B) Sources
38. Hydro-electric dams can be large sources of carbon dioxide and methane because:
    A) They use CO_2 to rotate their turbines

B) Gas escapes rapidly as water passes through them, forming CO_2 and CH_4.
C) Of the gas used in the process of maintenance of the dams
D) Of the influences of hydrostatic pressure
E) They are not sources of CO_2 or methane.

39. What type of lake is most likely to hold highly acidic water?
A) Volcanic Lakes
B) Fluvial Lakes
C) Glacial Lakes
D) Tectonic Lakes
E) Solution Lakes

40. Which of the following is the most geologically active region on Earth?
A) Glacial Valley
B) Mountain Peaks
C) Tectonic Regions
D) Recharge Zones
E) None of the above

41. What is the recurrence interval of a 100-year flash flood?
A) Typically 50 years
B) Typically 100 years
C) Typically 150 years
D) Typically 25 years
E) Typically 125 years

42. Which best describes the neck of a river?
A) point of land between the two edges of a meander
B) where the water going through a meanders drops alluvium on the inner bank
C) when the stream erodes through the neck causing the river to be back to a straight course
D) A separate body of water from a stream
E) None of the above

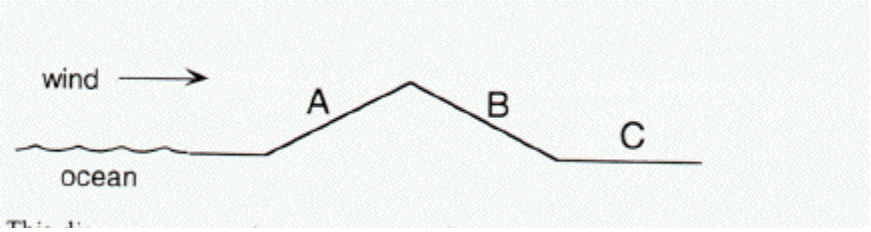
43. Viscous force is the _____ of shear stress due to viscosity and cross-section area of flow.
A) Sum
B) Difference
C) Product
D) Ratio
E) Square root

44. A flow in which the quantity of liquid flowing per second is constant, is called _____ flow.
A) Steady
B) Streamline

C) Balanced
D) Fixed
E) Anchored

45. The Hydraulic gradient line (H.G.L.) represents the sum of .....
A) Pressure head and kinetic head
B) Kinetic head and datum head
C) Pressure head and datum head
D) Kinetic Head and Pressure Head
E) Pressure head,kinetic head and datum head

46. What is the process by which water enters the small pore spaces between particles in soil or rocks?
A) Sublimation
B) Infiltration
C) Evaporation
D) Precipitation
E) Transpiration

47. 

The above diagram represents a coastal mountain range. Which of the regions will likely receive the most rainfall?
A) A
B) B
C) C
D) B and C
E) All receive same amounts of rainfall

48. Which of the following rocks has the highest permeability?
A) Uncemented Sandstone
B) Cemented SandStone
C) Unfractured Shale
D) Fractured Shale
E) All have approximately same permeability

49. The boundary between the saturated zone and the unsaturated zone of groundwater is called the _____.
A) Aquifer
B) Aquitude
C) Porosity
D) Water Table

E) Discharge Zone

50. In what type of rock do most caves form?
    A) Sandstone
    B) Shale
    C) Limestone
    D) Granite
    E) None of the above
51. Which of the following can contaminate an aquifer?
    A) Gasoline Station
    B) Agricultural Regions
    C) Landfills
    D) All of the Above
    E) None of these
52. What is the name of the region underground in which there is zero groundwater?
    A) Zone of Aeration
    B) Zone of Saturation
    C) Discharge Zone
    D) Recharge Zone
    E) Water Table
53. Where is the water table located in the event that the ground is saturated?
    A) Above ground level
    B) Below ground level
    C) At ground level
    D) Non existent
54. Give the equation for interstitial velocity, given that V = interstitial velocity, C stands for hydraulic conductivity, G is the hydraulic gradient and P is the medium's porosity.
    A) V= (CG)/P
    B) V = CGP
    C) V = C/(GP)
    D) V = P/(CG)
    E) V = G/(CP)

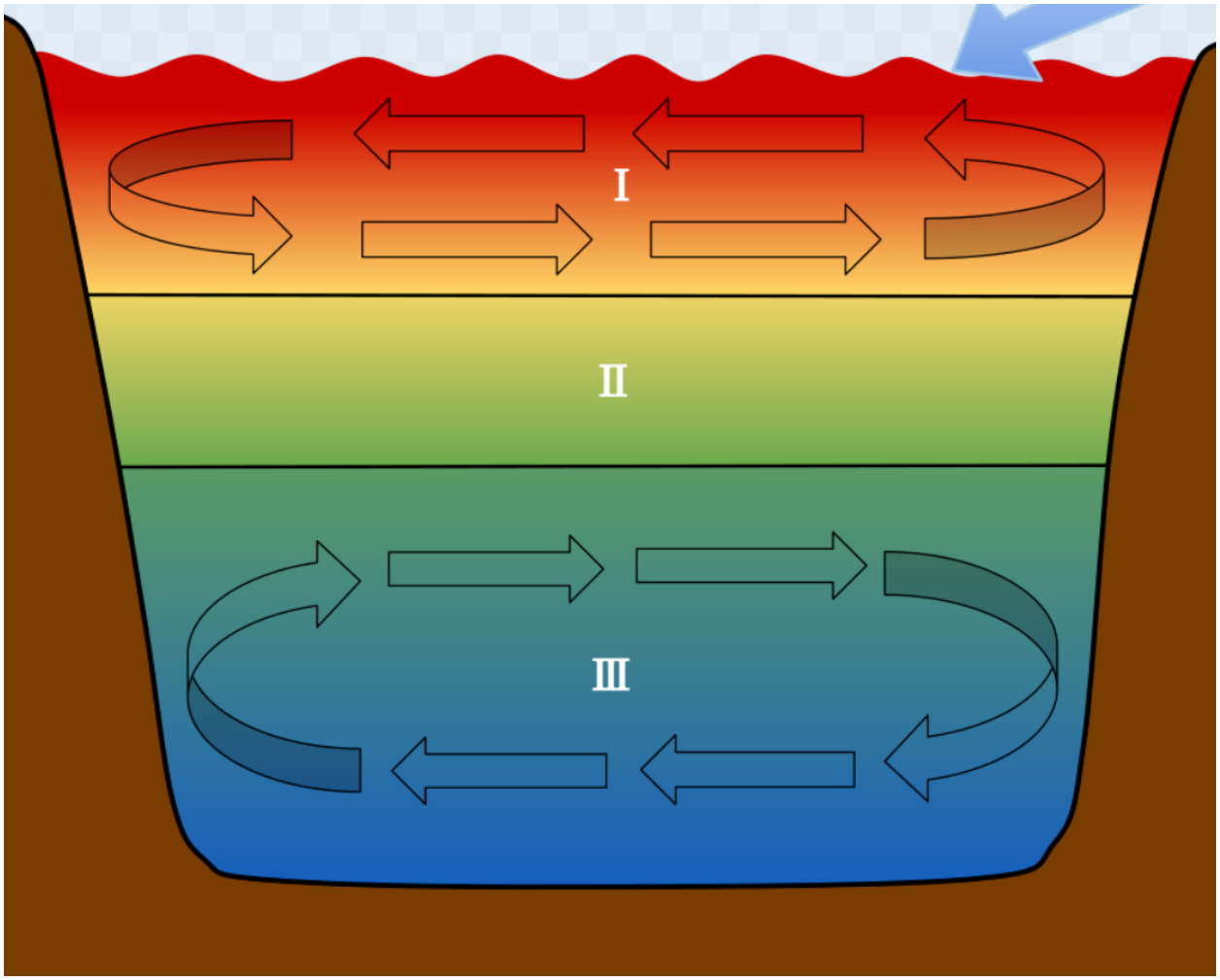
Questions 55 and 56 will be based on the following information. Assume the porosity of a given substance is n = 0.3, and the hydraulic conductivity is K = 100 ft/day.

55. What is the interstitial velocity of the water?
    A) 3.0 ft/day
    B) 3.1 ft/day
    C) 1.5 ft/day
    D) 1.15 ft/day
    E) 1.25 ft/day

56. How long will it take to travel 1 mile?
    A) 863 days
    B) 1715 days
    C) 1720 days
    D) 855 days
    E) 1725 days
57. What kind of aquifer is also commonly referred to as a water table aquifer?
    A) Ground Aquifer
    B) Confined Aquifer
    C) Unconfined Aquifer
    D) Above Ground Aquifer
    E) Below Grow Aquifer
58. What would be the pressure of the upper surface of an unconfined aquifer?
    A) Equal to atmospheric pressure
    B) Greater than atmospheric pressure
    C) Less than atmospheric pressure
    D) Very high pressure
    E) Very low pressure
59. TRUE OR FALSE: Perched water is a reliable source of water.
    A) False
    B) True
60. How many types are aquifers classified into based on physical conditions that water can exist in them?
    A) 1
    B) 2
    C) 3
    D) 4
    E) 5

Section 2 - Diagrams and Identification

1. Identify layers of Thermal Stratification (3 Points)



2. For each of the following, name the geographic feature corresponding to each of the images. (Figures A-P on Figure Sheet) (1 point each)

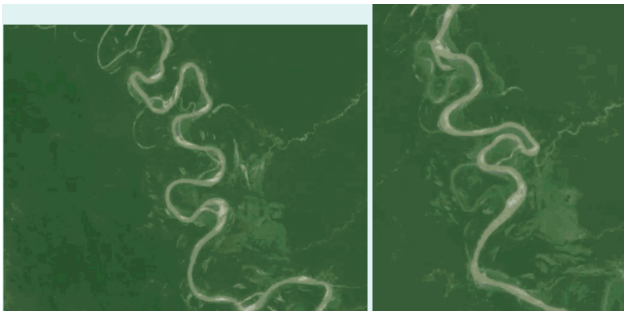
Section 3 – Short Answer/Free Response

Question 1 (3 points)

Find the approximate sinuosity of this river, using the given image:



Question 2 (8 points)



A) How is the sinuosity of these rivers changing over time?
B) Define a point bar. Which river is likely to have a higher magnitude of point bars? Why?
C) What is the hypothesized average of all rivers in the world's sinuosities?

Question 3 (14 points)

A) Define Darcy's Law and Write the equation. Identify all variables.
B) What is the difference between specific yield and specific retention?
C) If the distance between two wells is 12.5 meters, and the piezometric surface drop between two wells is 3.6 meters, find the time it would take for a molecule of water to move from one well to the other. Assume steady unidirectional flow in a homogenous sand confined aquifer with hydraulic conductivity of K = 0.125 m/day and an effective porosity of 0.5. Round to the tenth place for your final answer.

Question 4 (6 points)

A) What are some ways in which pollutants can move into the groundwater system more easily?
B) Explain the phenomenon of inter aquifer leakage.
C) Name at least 2 industrial contaminants that could impact aquifer systems.
D) What is the impact of aquifer pollution on human society?

Question 5 (7 points)

The following image shows the contours of a stream, with intervals of 5 meters each. The index contour, as shown in the image, has an elevation of 70 meters.



A) In which cardinal direction is this stream flowing in?
B) Calculate the gradient between the highest and lowest contour lines that cross the stream on this map. Assume the distance between the two is 3 meters.
C) What is the relief of this map?

Question 6 (7 points)

A) What type of karst is most likely to host a high number of hydrocarbons? What are the practical implications of this?
B) Define a kegel karst. What is unique about the structure of this kind of karst?
C) What is typical of the pH of soil around a karst?

Question 7 (2 points)

A) What is a human activity that causes a loss in sediment export and soil from natural environments?
B) What is the impact of surface runoff on soil? How is this important?